

Notes: Goeree (Econometrica, 2008)

Limited Information and Advertising in the U.S. Personal Computer Industry

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1 Big picture

- **Question.** How should researchers estimate demand, substitution, and market power when consumers are not aware of all products in the market?
- **Empirical setting.** The paper studies the U.S. home personal computer market from 1996 to 1998, a period with rapid product turnover, many new PC models, and large advertising expenditures.
- **Core friction.** Traditional BLP-style discrete-choice models assume that every consumer knows every product. Goeree relaxes this by letting each consumer have a stochastic product-specific choice set.
- **Role of advertising.** Advertising is modeled as informative: it affects the probability that a consumer is aware of a product, rather than simply shifting utility directly.
- **Heterogeneity.** The limited information model incorporates three forms of heterogeneity: tastes, choice sets, and media exposure. Different consumers see different media and therefore receive different product information.
- **Data combination.** The paper combines aggregate product-level PC sales and prices, aggregate advertising by media, and micro-level Simmons data on demographics, media exposure, and manufacturer choice.
- **Main result.** Full-information demand models make products look too substitutable and demand curves too elastic. Allowing for limited information raises estimated markups sharply.
- **Markup implication.** The median markup is about 15–19% under limited information but only about 5% under a full-information benchmark.
- **Competition implication.** Limited consumer awareness reduces the effective set of competitors facing a product. This lowers own-price elasticities and makes firms appear to have more market power.
- **Policy implication.** Merger analysis, market power measurement, and advertising policy can be misleading if researchers treat all consumers as fully informed.

2 Incremental contribution of the paper

- **Beyond BLP taste heterogeneity.** BLP shows that allowing for heterogeneous tastes is crucial for realistic elasticities. Goeree shows that allowing for heterogeneous *information sets* is also crucial in rapidly changing differentiated-product markets.
- **Choice-set formation inside structural demand.** The paper embeds stochastic consideration sets in a random-coefficients discrete-choice model, so consumers choose only among products they know.
- **Advertising as information, not taste.** Earlier empirical demand models often omit advertising or put it directly into utility. Goeree models advertising as changing product awareness probabilities, which is a different economic mechanism.
- **Data-combination method.** The paper shows how micro data on media exposure and demographics can discipline an advertising-information model even without individual-level advertising exposure and product purchase histories.
- **Media-specific advertising effects.** Advertising is allowed to work differently across magazine, newspaper, television, and radio channels, and those media effects differ across household demographics.
- **Implication for markups.** The paper demonstrates that full-information models can severely understate market power because they overstate the availability of substitutes to consumers.
- **Group advertising.** The model incorporates product-specific and group advertising, allowing a firm to advertise product families and internalize substitution among its own products.
- **Empirical IO contribution.** The paper provides one of the early structural empirical frameworks for estimating limited consumer information in a large differentiated-products market using aggregate and micro moments together.

3 Data and measurement

3.1 Product-level PC data

The product-level data come from Gartner and contain quarterly shipments and dollar sales of PCs sold from 1996 to 1998. A product is defined as a manufacturer–brand–CPU type–CPU speed–form factor combination. The paper focuses on home-market sales by 15 firms, which account for more than 80% of home PC sales on average.

- Product-level observations are model-quarter pairs.
- The analysis sample contains 2,112 model-quarter observations and 723 distinct models.
- Market size is the number of U.S. households.
- Market share is unit sales divided by market size.
- The outside option includes no purchase, used PCs, and purchases from firms not included among the 15 firms.

Interpretation: The narrow product definition is important because PCs are differentiated along manufacturer, brand, processor, speed, and form factor. The outside option is large, so the model must distinguish substitution across PCs from substitution into nonpurchase.

3.2 Advertising data

Advertising data come from Competitive Media Reporting. The raw data report quarterly advertising expenditures across many media, which the paper aggregates into four broad channels:

- magazines;
- newspapers;
- television;
- radio.

The data include product-specific and group advertising. For a product j , observed group advertising is converted into effective product advertising using:

$$ad_j = \sum_{H \in G_j} (\pi_1 ad_H + \pi_2 ad_H^2),$$

where G_j is the set of product groups that include product j .

Interpretation: Group advertising matters because PC firms often advertise product families rather than a single model. The model lets group advertising spill across products sold by the same firm and estimates whether there are economies of scope in such advertising.

3.3 Consumer-level data

Consumer-level information comes from the Simmons Survey of Media and Markets. The paper uses two survey years, 1996 and 1997, and draws a final sample of 13,400 respondents.

- Simmons reports household demographics, PC ownership, whether a PC was purchased recently, and manufacturer choice.

- Simmons also reports media habits, such as cable access, hours of television or radio, and newspaper or magazine readership.
- CPS data are used for additional population moments.

Interpretation: The consumer data do not observe product-level ad exposure and product-level purchases jointly. The key measurement step is therefore to use media-exposure patterns as micro moments that discipline how aggregate advertising maps into consumer awareness.

3.4 Main empirical variables

- **Market shares:** unit sales divided by U.S. households.
- **Prices:** dollar sales divided by unit sales, deflated by CPI.
- **Product characteristics:** CPU speed, Pentium indicator, laptop indicator, firm/manufacturer dummies, and product age.
- **Advertising:** effective product advertising by media and group advertising.
- **Media exposure:** household-level exposure to media channels from Simmons.
- **Markup:** $(p_j - mc_j)/p_j$, inferred from the estimated demand system and firm pricing first-order conditions.

Interpretation: The central measurement object is not advertising expenditure alone. It is the probability that a given consumer is informed about a given product as a function of advertising, product age, firm effects, and the consumer's media exposure.

4 Models and empirical specifications

4.1 Consumer utility

Consumer i 's indirect utility from product j in period t is:

$$u_{ijt} = \delta_{jt} + \mu_{ijt} + \varepsilon_{ijt},$$

where

$$\delta_{jt} = x'_j \beta + \xi_{jt}$$

contains mean utility and unobserved product quality. Taste heterogeneity is:

$$\mu_{ijt} = \alpha \ln(y_{it} - p_{jt}) + x'_j(\Omega D_{it} + \Sigma \nu_i), \quad \nu_i \sim N(0, I).$$

Interpretation: This is the random-coefficients demand core. Consumers differ in income, demographics, and unobserved tastes. The innovation is that a consumer cannot choose a product unless she is first informed about it.

4.2 Limited information and stochastic choice sets

Let ϕ_{ijt} be the probability that consumer i is informed about product j . Conditional on the random choice set, the purchase probability integrates over all choice sets that include j :

$$s_{ijt} = \sum_{S \in C_j} \left[\prod_{l \in S} \phi_{ilt} \prod_{k \notin S} (1 - \phi_{ikt}) \right] \frac{\exp(\delta_{jt} + \mu_{ijt})}{y_{it}^\alpha + \sum_{r \in S} \exp(\delta_{rt} + \mu_{irt})}.$$

Because there are many products, this expression is computationally infeasible if evaluated over all possible choice sets. The paper therefore simulates choice sets.

Interpretation: A product's effective competition depends on which other products are also known to the same consumer. This is why limited information changes substitution patterns, not just levels of demand.

4.3 Information technology

The probability of awareness follows a logit form:

$$\phi_{ij}(\theta_\phi) = \frac{\exp(\gamma_j + \lambda_{ij})}{1 + \exp(\gamma_j + \lambda_{ij})}.$$

The common product component is:

$$\gamma_j = a'_j(\varphi + \rho a_j + i_m \Psi_f) + \vartheta x_j^{age}.$$

The consumer-specific information component is:

$$\lambda_{ij} = a'_j(\Upsilon D_i^s \zeta + \kappa_i) + D'_i \lambda, \quad \ln \kappa_i \sim N(0, I).$$

Interpretation: Advertising, product age, firm advertising effectiveness, demographics, media exposure, and unobserved consumer heterogeneity jointly determine whether a product enters the consumer's consideration set.

4.4 Pricing and advertising first-order conditions

The firm sets prices according to multiproduct Bertrand first-order conditions:

$$s_j(p, a) + \sum_{r \in J_f} (p_r - mc_r) \frac{\partial s_r(p, a)}{\partial p_j} = 0.$$

Advertising medium choices satisfy:

$$M \sum_{r \in J_f} (p_r - mc_r) \frac{\partial s_r(p, a)}{\partial a_{jm}} + mr_j^{nh} = mc_{jm}^{ad}.$$

Interpretation: A firm advertises more when advertising raises profitable sales. Therefore advertising depends both on demand responsiveness and on markups. This is why markups and advertising decisions are tightly connected in the model.

4.5 Estimation and moments

The paper estimates the model using macro and micro moments:

- aggregate product market shares;
- micro moments linking demographics to manufacturer choice;
- micro moments linking demographics to media exposure;
- supply-side moments from pricing and advertising first-order conditions.

Interpretation: The data-combination strategy is essential because the paper lacks a single household dataset containing product choice, product-level ad exposure, and full advertising history.

5 Identification and empirical design

5.1 Main endogeneity problem

Prices and advertising are likely correlated with unobserved product quality ξ_j and cost unobservables ω_j . The identifying assumption is:

$$E[\xi_j(\Theta_0) \mid z] = E[\omega_j(\Theta_0) \mid z] = 0.$$

Interpretation: If high-quality products have high prices, naive demand estimates understate price sensitivity. If high-quality products receive more advertising, naive estimates overstate advertising effects. The paper therefore uses instruments.

5.2 Product and advertising instruments

The instruments are functions of:

- own and rival observed product characteristics;
- cost shifters;
- advertising cost shifters;
- characteristics of all competing products.

The logic follows BLP: rival product characteristics shift equilibrium markups and therefore prices. In this paper, they also shift optimal advertising because advertising depends on the profitability and substitutability of products.

Interpretation: The advertising instruments are an important extension. In a limited-information model, advertising is endogenous for the same reason prices are endogenous: high-quality or high-margin products may be advertised more.

5.3 Identification of the information technology

The parameters governing ϕ_{ij} are identified from several sources of variation:

- differences in advertising levels across products and media;
- differences in media exposure across household types;
- differences in purchase and manufacturer choice across demographics;
- product age variation, because older products have had more time to become known;
- firm advertising-effectiveness fixed effects.

Interpretation: The model does not infer limited information from low market shares alone. It uses consumer media exposure and demographic variation to separate awareness from taste.

5.4 Media exposure exogeneity

A concern is that consumers who plan to buy a PC may choose media that contain PC advertising. The paper tests whether media exposure is endogenous to the PC purchase decision using subscription prices and related media access costs as instruments.

Interpretation: The exogeneity tests support treating media exposure as predetermined in the structural model. This matters because media exposure is used to identify heterogeneity in information, not preference for PCs.

5.5 Specification and goodness-of-fit tests

The paper compares the limited information model to:

- a full-information BLP benchmark;
- a model where advertising directly shifts utility;
- a large-market-shares-only model related to Bajari and Benkard.

Interpretation: These comparisons ask whether the limited-information structure adds explanatory content beyond flexible utility, product unobservables, and standard random-coefficients substitution.

5.6 Limits to identification

- The model is static, so it does not estimate long-run brand-building effects of advertising.
- Product choice sets are simulated, not directly observed.
- Some product characteristics such as memory and hard disk are not included.
- Advertising data cannot perfectly isolate home-sector advertising from non-home PC or server advertising.
- Product characteristics are treated as exogenous.

Interpretation: The estimates are best read as a structural measurement of how limited awareness affects substitution and markups, not as a randomized causal experiment on advertising exposure.

6 Tables and figures

6.1 Table I: Summary Statistics for Market Shares, Advertising, Prices, and Markups; Table II: Descriptive Statistics for Simmons Data

Read:

- Table I shows that the PC market is concentrated. The top six firms account for more than 65% of dollar home-market share in each year and 75.26% in 1998.
- Top firms advertise much more heavily than the industry average: the top-six ad-to-sales ratio is 9.1%, versus 3.4% for the industry.
- HP and IBM have especially high advertising intensity, with HP at 17.7% and IBM at 20.1% ad-to-sales.
- Table I also previews the main markup theme: top firms have higher markups and higher advertising than the average firm.
- Table II shows that the Simmons sample closely matches the population on demographics, and that 46.6% of respondents own a PC while 11.3% recently bought one.
- The media exposure variables show large variation across households, which is used to identify heterogeneous ad exposure.

Economic message: The first pair of tables establishes the market environment: concentrated PC producers, large advertising outlays, and household-level variation in media exposure. This is the setting in which limited awareness can matter.

TABLE I
SUMMARY STATISTICS FOR MARKET SHARES, ADVERTISING, PRICES, AND MARKUPS^a

Manufacturer	Percentage Dollar Home Market Share			Average Annual		Median Percentage Markup, Home Sector	
	1996	1997	1998	Ad Expend	Ad-to-Sales Ratio	Median Price Home Sector	Over Marginal Costs Including Ad Costs
Industry					3.4%	\$2239	15%
Top 6 firms	65.67	68.31	75.26	\$469	9.1%	\$2172	17%
Acer	6.20	6.02	4.37	\$117	5.4%	\$1708	11%
Apple	6.66	5.79	9.16	\$161	5.3%	\$1859	16%
AST	3.08	1.53					13%
Compaq	11.89	16.29	16.43	\$208	2.4%	\$2070	23%
Dell	2.46	2.87	2.57	\$150	2.1%	\$2297	10%
Gateway	8.94	11.77	16.43	\$277	5.6%	\$2767	12%
Hewlett-Packard	4.02	5.52	10.05	\$651	17.7%	\$2203	16%
IBM	8.49	7.42	6.85	\$1189	20.1%	\$2565	16%
Micron	3.26	4.05	1.68				7%
NEC	23.48						
Packard-Bell							
Packard-Bell							
NEC		21.02	16.33	\$327	7.2%	\$2075	16%
Texas Instruments		1.40					7%
15 included	83.11	82.27	83.88				

^aNotes: Others in the 15 included are ATT (NCR), DEC, and Epson, each of which held less than 1% of the home (and total) market shares in 1996 and 1997. AST and Micron held less than 1% total market shares on average. In 1997 three mergers occurred: Packard Bell, NEC, ZDS; Acer, Texas Instruments; Gateway, Advanced Logic Research. Ad expenditures (in \$M) and ad-to-sales ratios are annual averages from LNA and include all sectors (home, business, education, government). Percentage markups are the median (price – marginal costs)/price across all products. The last column is percentage total markups per unit after including advertising. These are determined from estimated markups and estimated effective product advertising in the home sector.

(a) **Table I: Summary Statistics for Market Shares, Advertising, Prices, and Markups.** The table summarizes the market shares, advertising expenditures, ad-to-sales ratios, median prices, and markups by manufacturer. It shows the strong link between large incumbents, advertising intensity, and markups.

TABLE II
DESCRIPTIVE STATISTICS FOR SIMMONS DATA^a

Variable Description	Sample		Population	
	Mean	Std. Dev.	Mean	Std. Dev.
Male	0.663	0.474	0.661	0.473
White	0.881	0.324	0.881	0.324
Age (years)	47.38	15.68	46.87	15.13
30 to 50 (= 1 if 30 < age < 50)	0.443	0.497	0.449	0.497
Education (years)	13.98	2.54	14.00	2.35
Married	0.564	0.496	0.572	0.495
Household size	2.633	1.429	2.631	1.428
Employed	0.695	0.460	0.693	0.461
Income (\$)	56,745	45,246	56,340	44,465
Inclow (= 1 if income < \$60,000)	0.667	0.471	0.669	0.471
Inchigh (= 1 if income > \$100,000)	0.107	0.309	0.106	0.308
Own PC (= 1 if own a PC)	0.466	0.499	0.470	0.499
PCnew (= 1 if PC bought in last 12 months)	0.113	0.317	0.112	0.316
Media Exposure	Mean	Std. Dev.	Min	Max
Cable (= 1 if receive cable)	0.749	0.434	0	1
Hours cable (per week) cable (per week)	3.607	2.201	0	7
Hours noncable (per week)	3.003	2.105	0	6.2
Hours radio (per day)	2.554	2.244	0	6.5
Magazine (= 1 if read last quarter)	0.954	0.170	0	1
Number magazines (read last quarter)	6.870	6.141	0	95
Weekend newspaper (= 1 if read last quarter)	0.819	0.318	0	1
Weekday newspaper (= 1 if read last quarter)	0.574	0.346	0	1

^aNotes: Unless units are specified, variable is a dummy. Number of observations in survey is 39,931. Sample size is 13,400. Media exposure summary statistics are based on reports published by Simmons Market Research.

(b) **Table II: Descriptive Statistics for Simmons Data.** The table compares the Simmons estimation sample with the population and reports media exposure variables. These moments help discipline demographic and media-exposure heterogeneity in the information technology.

6.2 Table III: Structural Estimates of Utility and Cost Parameters

Read:

- CPU speed, Pentium status, laptop form factor, income-price utility, and firm fixed effects enter the utility side.
- CPU speed is valued positively and also interacts with household size, consistent with larger households valuing faster machines more.
- Laptop products have a positive mean utility coefficient but also large heterogeneity, reflecting the segmented nature of laptop demand.
- The cost side shows that CPU speed and laptop form factor raise marginal cost, as expected.
- The table jointly estimates demand and supply components, because markups and advertising decisions require marginal costs.

Economic message: Table III is the baseline structural demand and cost foundation. It shows that the model captures standard product differentiation before adding the central contribution: information-set heterogeneity.

MICHAEL C. CORTY, GUY CARL

TABLE III
STRUCTURAL ESTIMATES OF UTILITY AND COST PARAMETERS^a

Variable	Coefficient	Std. Error	Standard Deviation	Std. Error	Interactions With Demographics			
					Household size	Income > \$100,000	Age 30 to 50	White Male
Utility Coefficients								
Constant	-12.026**	(0.796)	0.044	(0.558)				
CPU speed (MHz)	9.288**	(1.599)	0.156**	(0.017)	4.049**			
					(0.674)			
Pentium	1.236*	(0.890)	0.209	(0.886)		0.016		
						(0.489)		
Laptop	2.974**	(0.525)	0.953	(4.619)			2.048	4.099
							(8.870)	(9.192)
ln(income – price)	1.211**	(0.057)						
Acer	2.624	(4.900)						
Apple	3.070**	(1.032)						
Compaq	2.662	(18.009)						
Dell	2.658**	(0.301)						
Gateway	7.411	(14.615)						
Hewlett–Packard	1.309	(3.905)						
IBM	2.514**	(0.712)						
Micron	-1.159	(6.011)						
Packard–Bell	4.372*	(4.002)						
Cost Side Parameters								
In marginal cost of production								
Constant	7.427**	(0.212)						
ln(CPU speed)	0.462**	(0.044)						
Pentium	-0.250**	(0.007)						
Laptop	1.204**	(0.071)						
Quarterly trend	-0.156**	(0.027)						
In marginal cost of advertising								
Constant	2.631	(7.087)						
Price of advertising	1.051**	(0.074)						
Non-Home Sector Marginal Revenue								
Constant	11.085	(278.374)						
Non-home sector price	1.815**	(0.354)						
CPU speed	0.010**	(0.004)						
Non-PC sales	3.688*	(1.881)						

^aNotes: ** indicates t -stat > 2; * indicates t -stat > 1. Standard errors are given in parentheses.

Table III: Structural Estimates of Utility and Cost Parameters. The table reports utility parameters, taste heterogeneity, demographic interactions, marginal production-cost parameters, marginal advertising-cost parameters, and non-home marginal revenue parameters. It is the structural backbone used to infer elasticities, markups, and counterfactuals.

6.3 Table IV: Structural Estimates of Information Technology Parameters

Read:

- The positive and significant media-exposure-by-advertising coefficient indicates that advertising exposure raises the probability that a consumer is informed about a product.
- Product age is positive, meaning older products are more likely to be known even absent current advertising.
- Television advertising and newspaper/magazine advertising have positive baseline effects but exhibit decreasing returns.
- Firm-level advertising effectiveness differs: Compaq, Dell, Gateway, IBM, and Packard-Bell ads are especially effective at informing consumers.
- Group advertising coefficients imply economies of scope, so it can be efficient to advertise product families rather than only individual models.

Economic message: Table IV is the core limited-information table. It shows that information is not randomly distributed across households: product awareness depends on media exposure, demographics, product age, media type, and firm advertising effectiveness.

TABLE IV
STRUCTURAL ESTIMATES OF INFORMATION TECHNOLOGY PARAMETERS^a

Variable	Coefficient Estimates for Interactions With Media							
	Magazine (mag)		Newspaper (np)		Television (TV)		Radio	
	Coefficient	Std. Error	Coefficient	Std. Error	Coefficient	Std. Error	Coefficient	Std. Error
Consumer Information Heterogeneity Coefficients								
Media and demographic interactions (Y)								
Constant	-1.052**	(0.040)	-0.973**	(0.040)	-1.032**	(0.041)	-1.000**	(0.043)
30 to 50 (= 1 if 30 < age < 50)	-0.042*	(0.025)	0.207**	(0.025)	0.019	(0.025)	-0.030*	(0.025)
50 plus (= 1 if age > 50)	0.005	(0.025)	0.541**	(0.025)	0.193**	(0.025)	-0.245**	(0.025)
Married (= 1 if married)	-0.022*	(0.018)	0.187**	(0.018)	0.075**	(0.018)	-0.011	(0.018)
hsize (household size)	0.040*	(0.006)	-0.038**	(0.006)	0.018*	(0.006)	0.012*	(0.006)
incow (= 1 if income < \$60,000)	-0.194**	(0.021)	-0.251**	(0.021)	0.114**	(0.021)	-0.117**	(0.022)
inhigh (= 1 if income > \$100,000)	0.153**	(0.029)	0.127**	(0.028)	-0.025	(0.030)	0.069**	(0.030)
malewh (= 1 if male and white)	-0.078*	(0.018)	0.002	(0.018)	-0.019*	(0.018)	0.006	(0.018)
eduhs (= 1 if highest edu 12 years)	-0.102*	(0.026)	-0.338**	(0.026)	0.296**	(0.027)	0.076**	(0.027)
eduad (= 1 if highest edu 1-3 college)	0.032*	(0.028)	-0.166**	(0.027)	0.278**	(0.028)	0.115**	(0.029)
edubs (= 1 if highest edu college grad)	-0.024	(0.025)	-0.063**	(0.024)	0.145**	(0.025)	0.081**	(0.026)
edsep (education if <11)	-0.028**	(0.003)	-0.069**	(0.003)	0.034**	(0.003)	-0.014**	(0.003)
Advertising media exposure (Z)								
media exposure* advertising	0.948**	(0.059)						
Demographics (A)								
Constant	0.104**	(0.004)						
High school graduate	0.834**	(0.028)						
Income < \$60,000	0.687**	(0.009)						
Income > \$100,000	0.139	(0.318)						

(Continues)

TABLE IV—Continued

Variable	Coefficient Estimates for Interactions With Media							
	Magazine (mag)		Newspaper (np)		Television (TV)		Radio	
	Coefficient	Std. Error	Coefficient	Std. Error	Coefficient	Std. Error	Coefficient	Std. Error
Information Technology Coefficients Common Across Consumers								
Age of PC	0.159**	(0.005)						
Media Advertising (ϕ, ρ)								
npand mag advertising	0.720*	(0.488)						
TV advertising	1.078**	(0.418)						
(np and mag advertising) ²	-0.013	(0.014)						
(TV advertising) ²	-0.049**	(0.004)						
Firm total advertising (Ψ)								
Acer	0.520	(0.042)						
Apple	0.163	(0.790)						
Compaq	0.504**	(0.077)						
Dell	0.497*	(0.460)						
Gateway	0.918**	(0.065)						
Hewlett-Packard	0.199	(11.750)						
IBM	0.926**	(0.184)						
Micron	0.029	(5.832)						
Packard-Bell	0.231*	(0.149)						
Group advertising (π)								
Group advertising	0.891**	(0.007)						
(Group advertising) ²	0.104**	(0.011)						

^aNotes: ** indicates t -stat > 2; * indicates t -stat > 1. Unless units are specified, variable is a dummy.

(a) **Table IV: Structural Estimates of Information Technology Parameters, part 1.** This part reports demographic-media interactions, the media-exposure effect of advertising, and baseline demographic information effects.

(b) **Table IV: Structural Estimates of Information Technology Parameters, part 2.** This continuation reports product age, media advertising effects, firm total-advertising effects, and group-advertising parameters.

6.4 Table V: A Sample from 1998 of Estimated Price and Advertising Elasticities

Read:

- Own-price elasticities are negative and finite; they are not as extremely elastic as in full-information models.
- Cross-price elasticities are small because many consumers are not aware of all substitute products.
- Advertising semi-elasticities show that a firm's advertising can raise awareness of one product while reducing rivals' market shares.
- Some advertising effects across products within a firm are less negative than effects on rival products, which helps explain group advertising.

Economic message: Table V shows the mechanism behind the paper's market-power result. If consumers have limited choice sets, even similar PCs may not be close substitutes for a given consumer, so demand curves are less elastic.

TABLE V
A SAMPLE FROM 1998 OF ESTIMATED PRICE AND ADVERTISING ELASTICITIES^a

	Apple PowerBook*	Apple Power Mac	Compaq Armada*	Compaq Presario	Dell Latitude*	HP Omnibook*	HP Pavilion	IBM PC	IBM Thinkpad*
Price Elasticities									
PowerBook*	-12.861	0.0692	0.0243	0.0287	0.0170	0.0219	0.0213	0.0182	0.0165
Power Mac	0.0856	-11.097	0.0202	0.0222	0.0196	0.0202	0.0248	0.0298	0.0364
Armada 7xxx*	0.0150	0.0107	-5.7066	0.0193	0.0606	0.0209	0.0203	0.0162	0.0426
Presario 2xxx	0.0122	0.0272	0.0125	-3.6032	0.0230	0.0272	0.0308	0.0348	0.0385
Latitude XPI*	0.0263	0.0274	0.0357	0.0261	-5.5701	0.0225	0.0217	0.0394	0.0453
Omnibook 4xxx*	0.0179	0.0147	0.0363	0.0298	0.0228	-5.6501	0.0269	0.0222	0.0499
Pavilion 6xxx	0.0118	0.0212	0.0153	0.0336	0.0167	0.0227	-5.1178	0.0396	0.0359
PC 3xxx	0.0137	0.0322	0.0137	0.0381	0.0153	0.0148	0.0325	-3.2626	0.0215
Thinkpad 7xxx*	0.0330	0.0192	0.0376	0.0195	0.0304	0.0425	0.0297	0.0291	-6.9745
Advertising Semielasticities									
PowerBook*	0.0076	-0.0057	-0.0142	-0.0110	-0.0044	-0.0139	-0.0166	-0.0072	-0.0097
Power Mac	-0.0057	0.0215	-0.0147	-0.0273	-0.0179	-0.0136	-0.0243	-0.0263	-0.0213
Armada 7xxx*	-0.0616	-0.0564	0.0017	-0.0057	-0.0314	-0.0625	-0.0441	-0.0684	-0.0948
Presario 2xxx	-0.0779	-0.0827	-0.0060	0.0120	-0.0208	-0.1092	-0.1413	-0.0825	-0.0830
Latitude XPI*	-0.0233	-0.0114	-0.0278	-0.0274	0.0230	-0.0380	-0.0239	-0.0199	-0.0438
Omnibook 4xxx*	-0.0034	-0.0042	-0.0039	-0.0043	-0.0064	0.0054	-0.0021	-0.0030	-0.0044
Pavilion 6xxx	-0.0036	-0.0045	-0.0038	-0.0082	-0.0051	-0.0066	0.0101	-0.0143	-0.0054
PC 3xxx	-0.0076	-0.0085	-0.0082	-0.0161	-0.0182	-0.0127	-0.0194	0.0095	-0.0029
Thinkpad 7xxx*	-0.0107	-0.0088	-0.0168	-0.0164	-0.0185	-0.0127	-0.0196	-0.0020	0.0089

^aNotes: A * indicates a laptop. For price elasticities, cell entries i, j , where i indexes row and j indexes column, give the percentage change in market share of brand i with a 1% change in the price of j . Each entry represents the median of the elasticities from 1998. For advertising elasticities, cell entries i, j give the percent change in the market share of i with a \$1000 increase in the advertising of j .

Table V: A Sample from 1998 of Estimated Price and Advertising Elasticities. The upper panel reports price elasticities; the lower panel reports advertising semi-elasticities. The table illustrates how limited awareness weakens cross-product substitution and makes advertising a product-specific information tool.

6.5 Table VI: Estimated Percentage Markups Under Limited and Full Information; Table VII: Median Product Price Elasticities Under Limited and Full Information

Read:

- Table VI compares implied markups under the limited-information model with a full-information benchmark.
- The industry median markup is 15% under limited information but 5% under full information, meaning the full-information model understates market power.
- For top firms and representative products, the markup gap is often very large because awareness limits effective competition.
- Table VII shows why markups differ: full-information own-price elasticities are much larger in absolute value than limited-information own-price elasticities.
- Full-information cross-price elasticities also imply more substitution than the limited-information model.

Economic message: The two tables jointly provide the central economic result: assuming full information makes the PC market look much more competitive than it is under limited awareness.

TABLE VI
ESTIMATED PERCENTAGE MARKUPS UNDER LIMITED AND FULL INFORMATION^a

	Median Percentage Markup		Char in Mar
	Under Limited Information	Under Full Information	
Total industry	15%	5%	67 ^c
Apple		2.5%	84 ^c
iMac	22.1%	3.1%	
Power Mac	13.7%	2.0%	
PowerBook [*]	10.0%	1.6%	
Compaq		7.0%	69 ^c
Armada 7xxx [*]	41.4%	3.5%	
Presario 2xxx	18.1%	2.6%	
Presario 1xxx [*]	15.2%	2.0%	
ProLinea	23.3%	7.0%	
Dell		1.8%	82 ^c
Latitude XPI [*]	7.0%	1.4%	
Dimension	15.5%	2.4%	
Inspiron	9.4%	1.6%	
Gateway		1.7%	86 ^c
Gateway Desk Series	12.8%	1.9%	
Gateway Portable Series	8.1%	1.5%	
HP		4.5%	72 ^c
OmniBook 4xxx [*]	8.3%	5.7%	
Pavilion 6xxx	22.7%	3.1%	
Vectra 5xx	15.8%	6.8%	
IBM		2.0%	88 ^c
Aptiva	16.0%	2.3%	
Thinkpad 7xxx [*]	7.4%	1.6%	
IBM PC 3xx	26.1%	2.1%	
Packard-Bell		3.0%	81 ^c
NEC Versa [*]	11.1%	1.6%	
NEC Desk Series	17.6%	2.5%	

^aNotes: Percentage markups are defined as (price – marginal cost)/price. Full information is the traditional π in which consumers know all products; under limited information the choice set is estimated. ^{*} indicates that con

TABLE VII
MEDIAN PRODUCT PRICE ELASTICITIES UNDER LIMITED AND FULL INFORMATION^a

	Apple Performa	Apple PowerBook [*]	Compaq Contura [*]	Compaq Presario	Dell Latitude [*]	Gateway Desk	Gateway Portable [*]	HP Pavilion	HP Vectra	IBM PC	IBM Thinkpad [*]	Pack-Bell Desk
Under Limited Information												
Performa	-8.119	0.085	0.018	0.034	0.013	0.009	0.006		0.021	0.019		0.023
PowerBook Duo [*]	0.061	-11.568	0.024	0.023	0.009	0.007	0.018		0.018	0.012		0.028
Contura [*]	0.014	0.010	-8.929		0.052	0.031	0.040		0.012	0.013		0.025
Presario 4xxx	0.011	0.011		-3.508		0.009	0.009		0.026	0.025	0.036	0.024
Latitude [*]	0.027	0.009	0.025		-8.344	0.042	0.046		0.011	0.009		0.033
Gateway Desk Series	0.015	0.008	0.034	0.105	0.008	-3.955	0.008	0.030	0.014	0.012	0.006	0.027
Gateway Portable Series [*]	0.029	0.015	0.055	0.037	0.013	0.015	-6.757	0.018	0.022	0.020	0.079	0.015
Pavilion 4xxx						0.133	0.026	-5.173			0.045	
Vectra XU	0.016	0.012	0.011	0.018	0.012	0.010	0.010		-5.534	0.026		0.036
IBM PC 7xx	0.013	0.007	0.029	0.019	0.007	0.008	0.037		0.007	-3.687		0.086
Thinkpad 6xx [*]				0.010		0.026	0.080	0.024			-5.209	
Packard-Bell Desk Series	0.008	0.005	0.003	0.018	0.007	0.006	0.004		0.012	0.022		-3.317
Under Full Information (BLP benchmark)												
Performa	-28.648	0.106	0.088	0.060	0.072	0.066	0.051		0.097	0.090		0.057
PowerBook Duo [*]	0.089	-31.654	0.047	0.099	0.060	0.058	0.046		0.060	0.076		0.060
Contura [*]	0.065	0.080	-31.721		0.235	0.307	0.128		0.050	0.038		0.028
Presario 4xxx	0.025	0.013		-29.491		0.038	0.099		0.131	0.128	0.062	0.061
Latitude [*]	0.030	0.010	0.160		-29.547	0.195	0.175		0.025	0.092		0.076
Gateway Desk Series	0.033	0.039	0.170	0.263	0.019	-34.213	0.011	0.107	0.011	0.012	0.038	0.069
Gateway Portable Series [*]	0.030	0.032	0.315	0.212	0.023	0.017	-34.453	0.133	0.023	0.023	0.060	0.017
Pavilion 4xxx						0.135	0.019	-35.362			0.090	
Vectra XU	0.069	0.040	0.031	0.017	0.080	0.080	0.047		-39.009	0.011		0.035
IBM PC 7xx	0.149	0.138	0.180	0.236	0.060	0.081	0.078		0.030	-20.780		0.209
Thinkpad 6xx [*]				0.163		0.080	0.056	0.069			-39.809	
Packard-Bell Desk Series	0.028	0.031	0.185	0.213	0.050	0.048	0.045		0.300	0.260		-26.327

^aNotes: A ^{*} indicates a laptop. For price elasticities, cell entries ϵ_{ij} , where i indexes row and j indexes column, give the percentage change in market share of brand i with a 1% change in the price of j . Each entry represents the median of the product elasticities over all quarters during which the PC was sold. The BLP benchmark is the BLP model with micromoments.

(a) **Table VI: Estimated Percentage Markups Under Limited and Full Information.** The table translates demand estimates into markups and shows that limited information raises implied market power.

(b) **Table VII: Median Product Price Elasticities Under Limited and Full Information.** The table compares own- and cross-price elasticities under both information structures and shows that full information makes products appear much more substitutable.

6.6 Table VIII: Goodness of Fit

Read:

- Table VIII compares observed market shares and group-advertising thresholds across models.
- The limited-information model predicts average annual unit market shares close to observed shares for major firms.
- The limited-information model has fewer group-advertising threshold misses than the uninformative-advertising model.
- The table also contrasts the role of unobserved product attributes across LIM, BLP, and other alternatives.

Economic message: Table VIII shows that the limited-information model does not achieve higher markups by sacrificing basic fit. It captures important dimensions of market shares and advertising behavior while changing the interpretation of substitution and competition.

TABLE VIII
GOODNESS OF FIT^a

		Prediction for Different Models			
Response Variable	Observed	Limited Information	Full Information Uninformative Advertising	Full Information No Advertising BLP	Large Market Shares Only Bajari-Benkard
Average Annual Percent Unit Market Shares					
Apple	6.45%	8.87%	8.96%	5.15%	6.54%**
Compaq	16.17%	17.75%*	17.98%	19.74%	22.16%
Gateway	10.76%	11.32%**	10.99%**	13.07%	13.34%
HP	6.53%	6.86%**	5.99%*	1.98%	7.85%
IBM	7.60%	8.51%*	8.59%	9.38%	8.10%*
Packard-Bell	22.61%	20.37%*	24.34%*	27.41%	27.00%
Mean Industry Elasticity		4.39%	4.41%	4.41%	4.38%
Group and Product-Specific Advertising					
Predicted threshold value (\$millions)		1.05	1.66	NA	NA
Percent group expenditures below predicted threshold value					
All products		2.7%	8.2%		
Apple		2.4%	8.2%		
Compaq		1.4%	4.4%		
Gateway		1.1%	2.6%		
HP		0.0%	0.0%		
IBM		1.1%	3.8%		
Packard-Bell		0.0%	0.0%		
Newspaper		0.0%	0.8%		
Magazine		0.1%	0.9%		
Television		1.5%	5.5%		
Radio		0.9%	0.9%		
Percent product-specific expenditures above predicted threshold value					
All products		0.8%	0.8%		
Apple		0.9%	0.9%		
Compaq		0.9%	0.9%		
Gateway		0.0%	0.0%		
HP		0.0%	0.0%		
IBM		0.0%	0.0%		
Packard-Bell		0.0%	0.0%		
Newspaper		0.0%	0.0%		
Magazine		0.8%	0.8%		
Television		0.1%	0.1%		
Radio		0.8%	0.8%		

^aNotes: Predicted market shares are evaluated at parameter estimates with unobserved product attributes restricted to zero. ** indicates predicted values within 5% of the observed value; * indicates within 10% of the true value. Predicted group advertising expenditures threshold value in millions. Advertising expenditures are computed using equation (1) evaluated at the optimal parameter values. Firm percentages are calculated as percent of product/medium advertising by that firm. The BLP model includes micromoments. The Bajari-Benkard (BB) model includes only those products which sold more than 5000 units. NA denotes not applicable.

Table VIII: Goodness of Fit. The table reports observed and predicted market-share statistics and advertising-threshold diagnostics for several models. It supports the claim that modeling awareness improves economic interpretation without producing a poor empirical fit.

7 Short summary

- The paper extends random-coefficients demand estimation by allowing consumers to know only a subset of products.
- Advertising affects the probability that consumers are informed about products.
- The model uses aggregate sales and advertising data plus Simmons micro data on demographics, media exposure, and manufacturer choice.
- The PC market is a natural setting because products turn over rapidly and firms advertise heavily.
- The information technology depends on media channel, product age, firm advertising effectiveness, demographics, and media exposure.

- Full-information models overstate substitution because they assume every consumer can switch to every product.
- The limited-information model yields less elastic demand and substantially higher markups.
- Top firms benefit most because they advertise more and are more effective at informing consumers.
- Group advertising is meaningful because ads can inform consumers about several related products.
- The paper’s main lesson is that consumer information heterogeneity can be as important as taste heterogeneity for estimating competition.

8 Conclusion

8.1 Core conclusion

Goeree shows that assuming full information in differentiated-products demand estimation can materially distort estimates of demand elasticities, substitution patterns, markups, and the intensity of competition. In the U.S. personal computer market, consumers appear to know only a subset of products, and advertising is an important mechanism through which firms place products into consumers’ choice sets.

8.2 Incremental contribution relative to prior demand models

The paper’s central incremental contribution is to move from heterogeneous tastes alone to heterogeneous information sets. BLP-style demand models allow consumers to value products differently, but they usually assume that every consumer is aware of every product. Goeree shows that this assumption is untenable in markets with rapid product turnover and heavy advertising. The paper also shows how to use micro media-exposure data to identify awareness heterogeneity when individual product-level advertising exposure data are unavailable.

8.3 Economic intuition

A product competes only with products consumers know. If a consumer is unaware of a rival, that rival is not in the consumer’s effective choice set, even if it exists in the market. Advertising therefore affects competition by changing awareness, not merely by changing utility. This makes demand less elastic and raises markups. The full-information model mistakes limited awareness for weak preference substitution or product unobservables, producing misleading elasticities.

8.4 Implications for market power and advertising policy

The results suggest that advertising can protect market power by widening information gaps. High-advertising firms inform more consumers and face fewer effective substitutes among consumers who are aware of their products. This matters for merger policy and antitrust analysis: a full-information demand model can understate market power and overstate competitive pressure when many consumers are unaware of alternatives.

8.5 Limitations and future directions

The model is static and therefore abstracts from long-run brand-building effects, repeated search, learning over time, and dynamic product introductions. Choice sets are simulated rather than observed directly. Advertising data are aggregate, and product characteristics are incomplete. Future work could combine household-level purchase, product-level ad exposure, search behavior, and dynamic learning to separate informative advertising, persuasive advertising, and brand capital more fully.

8.6 Takeaway

The paper's enduring message is simple: observed product availability is not the same as consumer awareness. In rapidly changing markets, demand estimation should account for the fact that consumers may not know all products. Otherwise, researchers may conclude that markets are more competitive, demand is more elastic, and markups are lower than they really are.